

GROUP 33

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Case Study - 9

The Oil Mining and Usage Database helps manage oil extraction operations by keeping all important details in one place. It tracks data related to each oil well, such as location, depth, and status, and monitors production metrics like the volume of oil and gas extracted. The database also keeps records of equipment used, maintenance schedules, and any repairs needed to ensure everything runs smoothly. By providing a comprehensive overview of oil production activities, the database helps ensure resources are used efficiently and safely.

Additionally, the database supports financial management by tracking costs and budgets, which helps avoid overspending. It also monitors environmental impact and ensures compliance with safety regulations through regular inspections and assessments. This comprehensive approach improves decision-making and coordination among all parties involved in oil extraction, leading to a safer, more efficient, and sustainable operation.

Problem Overview

Oil extraction is one of the most complex and high-stakes industries in the world, with significant economic, environmental, and safety considerations. Managing the vast amount of data involved in oil extraction operations requires sophisticated tools and systems, which is where an Oil Mining and Usage Database comes into play. This database serves as a centralized hub for tracking and monitoring key metrics related to oil wells, equipment, production volumes, financial costs, environmental impact, and regulatory compliance.

In this document, we will examine the core issues involved in managing oil extraction operations and the challenges faced by companies in keeping track of all vital information. We will explore the specific problem areas, including the complexity of tracking multiple types of data, ensuring safety and compliance, and managing costs efficiently. This comprehensive description will focus solely on the problems involved and the importance of effective data management for the oil industry.

Oil mining or oil extraction is a process of extracting oil from **underground reservoirs** and India has **23 oil refineries** across the country.

First step : Locating oil reservoirs

To locate oil reservoirs **seismic surveys** and other methods are used. Methods like “**classic**” methods are used which include an **underground explosion** and observe the seismic response. **The effect of seismic ground vibrations on a structure**, such as a building is called seismic response. Other instruments like **magnetometers** are also used to measure magnetic field or magnetic dipole moment.

Every well in an oil field is **unique**, and companies need accurate records of their **location, depth, and extraction rates**. Without this information being properly recorded, companies may experience **inefficiencies in oil production and costly errors**. There are also concerns about **wells that are no longer operational**, which still need to be tracked to avoid regulatory issues.

Reference

1. ([https://www.investopedia.com/terms/e/estimated-ultimate-recovery.asp#:~:text=Estimated%20ultimate%20recovery%20\(EUR\)%20is,in%20concept%20to%20recoverable%20reserves.](https://www.investopedia.com/terms/e/estimated-ultimate-recovery.asp#:~:text=Estimated%20ultimate%20recovery%20(EUR)%20is,in%20concept%20to%20recoverable%20reserves.))
2. <https://indianpetroplus.com/wells-database>

Second Step : Extraction and Drilling

The oil well is created by drilling a long hole into the earth with an oil rig (machine that creates big holes deep down the earth surface). A steel pipe is placed in the hole, to provide structural integrity to the newly drilled well bore. Holes are then made in the base of the well to enable oil to pass into the bore. Finally, a collection of valves called a " is fitted to the top; the valves regulate pressures and control flow.

Average depths of oil wells are ranging from 5,000 feet to 20,000 feet.

Recovery of a well cannot be known with certainty until the well ceases production, petroleum engineers often determine an **estimated ultimate recovery (EUR)** based on **decline rate projections** years into the future. Various models, mathematical techniques, and approximations are used.

What is EUR ?

Estimated ultimate recovery is an **approximation of the quantity** of oil or gas that is potentially recoverable or has already been recovered from a reserve or well.

Reference

([https://www.investopedia.com/terms/e/estimated-ultimate-recovery.asp#:~:text=Estimated%20ultimate%20recovery%20\(EUR\)%20is,in%20concept%20to%20recoverable%20reserves.](https://www.investopedia.com/terms/e/estimated-ultimate-recovery.asp#:~:text=Estimated%20ultimate%20recovery%20(EUR)%20is,in%20concept%20to%20recoverable%20reserves.))

There are different recovery stages such as

- 1) Primary Stage
- 2) Secondary Stage
- 3) Enhanced Stage

- **Primary Stage**

When the underground pressure in the oil reservoir is sufficient to force the oil to the surface, all that is necessary to capture oil is to place a complex arrangement of **valves** on the wellhead and further to connect the well to a pipeline network for storage and processing. Sometimes, during primary recovery, to increase extraction rates, pumps, such as **beam pumps** and electrical **submersible pumps** (ESPs), are used to bring the oil to the surface; these are known as artificial lifting mechanisms.

Recovery factor during the primary recovery stage is typically **5-15%**

- **Secondary Stage**

Over the lifetime of a well, the pressure falls. After natural reservoir drive diminishes and there is insufficient underground pressure to **force the oil to the surface**, secondary recovery methods are applied. These rely on supplying external energy to the reservoir by **injecting fluids** to increase reservoir pressure, hence increasing or replacing the **natural reservoir drive with an artificial drive**. Secondary recovery techniques increase the reservoir's pressure by water injection, gas reinjection and gas lift.

The recovery factor after primary and secondary oil recovery operations is between **35 and 45%**.

- **Enhanced recovery**

Enhanced, or tertiary oil recovery methods, further increase mobility of the oil in order to increase extraction.

Thermally enhanced oil recovery methods (TEOR) are tertiary recovery techniques that heat the oil, reducing its viscosity and making it easier to extract. Steam injection is the most common form of TEOR, and it is often done with a **cogeneration plant**. This type of cogeneration plant uses a **gas turbine** to generate electricity, and the waste heat is used to produce steam, which is then injected into the reservoir

Tertiary recovery allows **another 5% to 15%** of the reservoir's oil to be recovered.

Reference

:
https://en.wikipedia.org/wiki/Extraction_of_petroleum#:~:text=Seismic%20surveys%20and%20other%20methods,such%20as%20tires%20and%20refrigerators.

Types of Oil :

- 1) Mineral Oil :** It is directly derived from refined crude petroleum oil with any unwanted hydrocarbons and natural contaminants removed during the process.
- 2) Synthetic Oil :** It is a base oil made artificially through a chemical process and is less thick than a mineral oil.
- 3) Semi-synthetic oil :** It is a mixture of mineral oil and synthetic oil and it is more expensive than mineral oil, but cheaper than synthetic oil.

Reference

<https://www.kandmtyres.co.uk/mineral-oil-and-synthetic-oil-explained/>

Third Step : Equipment Monitoring and Maintenance

Oil extraction relies on a complex array of machinery and equipment, from **drilling rigs to pipelines and pumps**. Monitoring the status, maintenance schedules, and repair histories of this equipment is vital to ensuring smooth operations. However, managing this data can be incredibly difficult without a reliable system in place.

Tracking Equipment Usage: Every piece of equipment has a unique life cycle, from **installation to eventual decommissioning**. Regular maintenance is required to keep machinery functioning efficiently and safely. **A major problem for oil companies is ensuring that they keep accurate records of equipment usage, maintenance needs, and repairs.**

Maintenance Scheduling: Companies often struggle with managing maintenance schedules for multiple pieces of equipment across different oil wells. Without a centralized system, there's a higher risk of equipment failure due to missed maintenance appointments, which can lead to costly repairs, production delays, or even accidents.

Why is Predictive Maintenance so Important in the Oil and Gas Industry?

An average oil and gas **company goes through at least 27 days of unplanned downtime each year**, costing **\$38 million**. Even if the downtime lasts for just **3.65 days**, the resulting losses can be as high as **\$5 million**.

That's why predictive maintenance is so important. Sophisticated predictive maintenance technologies **use artificial intelligence**, machine learning, and advanced analytics to spot issues and alert the relevant technicians, preventing potential equipment failure and safety risks. **According to a McKinsey report**, an offshore oil and gas company used a predictive maintenance solution to **reduce downtime by 20%**, leading to a **production increase of more than 500,000 oil barrels annually**.

Reference

<https://www.birlasoft.com/articles/predictive-maintenance-in-oil-gas-Industry>

Oil and Gas Predictive Maintenance Use Cases

Some predictive maintenance use cases in the oil and gas industry — from exploration and production to storage and processing.

- [Oil & Gas Pump Condition Monitoring](#)
- [Oil & Gas Vessel Maintenance & Monitoring](#)
- [Virtual Rig Monitoring](#)
- [Tank Pressure Monitoring](#)
- [Machinery Condition Monitoring System and Asset Protection](#)
- [Plant Performance Monitoring](#)
- [Pipeline Monitoring](#)
- [Drill Corrosion Detection and Maintenance](#)

Fourth Step : Ensuring Safety and Regulatory Compliance

Oil and gas companies are required to comply with strict regulatory requirements **related to EHS**. These regulations are designed to ensure that companies **operate safely, minimize environmental impacts, and protect workers**. Companies must **comply with local, state, and federal laws, as well as industry-specific regulations**.

Failing to comply with these regulations can result in **hefty fines**, environmental damage, or, worse, loss of life. As such, ensuring regulatory compliance through regular inspections, assessments, and **record-keeping is essential**.

Safety Inspections: Oil extraction sites are dangerous environments that require regular safety checks. Tracking the results of these inspections and ensuring follow-up actions are completed is a significant challenge. **Companies often need to maintain large amounts of documentation for regulatory purposes, and the absence of proper records can lead to non-compliance issues.**

Environmental Impact Monitoring: Oil extraction has a major impact on the environment, from the risk of spills to the emission of greenhouse gasses. As part of regulatory compliance, companies must monitor and report their environmental impact. **This involves keeping detailed records of air quality, water pollution,**

waste management, and hazardous materials which can be a complex task.

Reference

<https://nanoschool.in/nstc-blog/ehs-in-oil-gas-industries-best-practices-for-ensuring-safety-and-compliance/#:~:text=Oil%20and%20gas%20companies%20must,safety%20training%2C%20and%20safety%20audits.>

Fifth Step: Coordination Between Multiple Stakeholders

Oil extraction **involves coordination between a variety of parties, including engineers, financial analysts, environmental specialists, and regulatory bodies.** Effective communication and coordination between these stakeholders are crucial for the success of an oil extraction operation. They influence public perception, regulatory approval, and financial support.

Data Sharing

Different stakeholders need access to different types of data, whether it's production metrics for engineers or financial reports for analysts. Ensuring that everyone has access to the right data at the right time can be a major challenge for oil companies, especially when they rely on outdated or disconnected systems.

Decision-Making Based on Incomplete Information: Without a centralized system that brings all the data together, decision-makers in the oil industry may struggle to make informed choices. For example, an operations manager may need to decide whether to halt production for equipment maintenance but may not have access to accurate production forecasts or cost estimates.

Reference

1. <https://www.boreal-is.com/blog/oil-and-gas-stakeholders/>
2. https://www.researchgate.net/publication/379040142_EFFECTIVE_STAKEHOLDER_RELATIONSHIP_MANAGEMENT_IN_THE_OIL_GAS_SECTOR_A_CONCEPTUAL_AND_REVIEW_PERSPECTIVE

Sixth Step: Financial Management Challenges

Oil extraction is capital-intensive, with high costs related to drilling, equipment, and operations. To ensure profitability, **companies must closely monitor their budgets and avoid overspending**. However, financial management in this industry is particularly challenging due to the number of variables involved.

Tracking Costs and Budgets: Oil companies face the challenge of tracking all costs associated with each oil well and extraction operation. This includes everything from the cost of drilling to equipment maintenance and labor. Without accurate financial data, companies risk overspending and decreasing their profit margins.

Another angle to consider is that as **old wells rapidly dry up**, the need for **unconventional exploration** such as **horizontal drilling** and **ultra-deepwater drilling** is increasing. Agreed, these techniques **provide avenues for more revenue but the cutting-edge technology required comes at higher costs, risks, and complexity** than traditional drilling.

Reference

[https://blog.isa.org/maintenance-management-increase-productivity-minimize-production-costs-oil-gas-industry#:~:text=3\)%20Asset%20Maintenance.%20Oil%20in%20industry%20assets%20are,\(fixed%20or%20rotating\)%2C%20facilities%2C%20oil%20fields%2C%20rigs%2C%20etc.](https://blog.isa.org/maintenance-management-increase-productivity-minimize-production-costs-oil-gas-industry#:~:text=3)%20Asset%20Maintenance.%20Oil%20in%20industry%20assets%20are,(fixed%20or%20rotating)%2C%20facilities%2C%20oil%20fields%2C%20rigs%2C%20etc.)

Asset Maintenance: Equipment failure can lead to unplanned expenses, and oil companies need to allocate budgetary resources for maintenance and repairs. However, companies often struggle to predict these costs accurately due to the unpredictable nature of equipment failures and repairs.

Reference

[https://blog.isa.org/maintenance-management-increase-productivity-minimize-production-costs-oil-gas-industry#:~:text=3\)%20Asset%20Maintenance.%20Oil%20in%20industry%20assets%20are,\(fixed%20or%20rotating\)%2C%20facilities%2C%20oil%20fields%2C%20rigs%2C%20etc.](https://blog.isa.org/maintenance-management-increase-productivity-minimize-production-costs-oil-gas-industry#:~:text=3)%20Asset%20Maintenance.%20Oil%20in%20industry%20assets%20are,(fixed%20or%20rotating)%2C%20facilities%2C%20oil%20fields%2C%20rigs%2C%20etc.)

Seventh Step: Ethical Considerations

Public and Investor Scrutiny: Oil companies are increasingly under pressure from investors and the public to minimize their environmental impact and adopt more sustainable practices. This requires **accurate data on their operations and environmental metrics**, which can be difficult to collect and manage.

Ethical Accountability: In addition to environmental concerns, oil companies must also ensure that they are operating ethically, especially in regions where labor practices and working conditions may be a concern.

Key ethical issues currently facing the industry include:

- **Environmental Impact:**
The extraction, production, and consumption of fossil fuels contribute significantly to greenhouse gas emissions and climate change. **The industry is under pressure to reduce its carbon footprint** and invest in renewable energy sources.
- **Corporate Governance:**
There are concerns about **transparency, accountability, and corruption within the industry**. Effective governance practices are essential to ensure ethical decision-making and compliance with regulations.
- **Social Responsibility:**
Companies are expected to contribute positively to the communities in which they operate, including respecting human rights, providing fair wages, and supporting local development.
- **Safety and Health:**
Ensuring the safety and health of workers is a critical ethical issue, particularly given the hazardous nature of oil and gas operations.

Reference

<https://www.linkedin.com/pulse/ethical-considerations-oil-gas-industry-explained-deepak-rastogi-uj6vc/>

Data Requirements for Constructing the database for Oil Mining and Usage.

Based on the problem areas described, the data needed to construct an effective Oil Mining and Usage Database would include:

Oil Well Data:

Location (latitude, longitude)

Depth and dimensions

Operational status (active/inactive)

Daily/weekly/monthly production metrics (oil and gas volumes)

Equipment Data:

Equipment type and ID

Installation date and expected life cycle

Maintenance schedules

Repair history and costs

Safety and Compliance Data:

Safety inspection records

Compliance reports for environmental regulations

Incident reports (e.g., spills, leaks, accidents)

Financial Data:

Drilling and operational costs

Maintenance and repair costs

Budget allocations and spending reports

Environmental Data:

Greenhouse gas emissions data

Oil spill/leak records

Impact assessments for environmental compliance

Stakeholder Coordination Data:

Communication logs

Reports shared between departments

Decision-making records and outcomes